



Some Properties of Prime-Cubed Cyclotomic Cosets in Cyclic Groups

Panthakan Boonsuriyatham¹, Somphong Jitman^{1*}

¹Department of Mathematics, Faculty of Science, Silpakorn University
Email: panth.b@hotmail.com



ABSTRACT

Cyclotomic cosets in cyclic and abelian groups have been of interest and extensively studied due to their rich algebraic structures and wide applications. In this project, we focus on q -cyclotomic cosets of the cyclic group $\mathbb{Z}_m$ in the case where $q = p^3$ is a prime-cubed integer and m is a positive integer such that $\gcd(m, q) = 1$. Here, 6 types of cyclotomic cosets in $\mathbb{Z}_m$ are introduced and studied. Properties, characterizations and enumerations of such q -cyclotomic cosets are given.

INTRODUCTION

In [1], self-dual cyclic codes of length n over $\mathbb{F}_q$ have been studied based on q -cyclotomic cosets in $\mathbb{Z}_n$ when n is odd and $q = 2^m$ for some positive integer m . Characterization and enumeration of Euclidean self-dual and Euclidean self-orthogonal abelian codes are studied in [2]. Later, Hermitian self-dual quasi-abelian codes are studied in [3]. They are all related to q -cyclotomic class of some types. In [4], formulas for the number of 2^l -cyclotomic and 2^{2l} -cyclotomic cosets of some types in the cyclic group $\mathbb{Z}_{p^r}$ are simplified.

From explanations above, some types of q -cyclotomic cosets where $q = 2^m$ have been long considered. Therefore, it is interesting to study q -cyclotomic of other types where $q = p^m$ for any prime p and a positive integer m . In this research, we consider $m = 3$ and introduce other types of q -cyclotomic cosets. Precisely, q is a **prime-cubed**. Properties, characterizations and enumerations of such q -cyclotomic cosets will be studied.

DEFINITIONS

Definition 1 For each $a \in \mathbb{Z}_m$ and prime power q such that $\gcd(q, m) = 1$, the q -**cyclotomic coset** of $\mathbb{Z}_m$ **containing** a is defined to be

$$S_q(a) := \{q^i \cdot a \mid i = 0, 1, 2, \dots\}, \text{ where } q^i \cdot a := \sum_{j=1}^{q^i} a \text{ in } \mathbb{Z}_m.$$

For convenience, denote by $q^i a$ the element $q^i \cdot a \in \mathbb{Z}_m$.

Definition 2 Let $a \in \mathbb{Z}_m$ and q be a prime power. A q -cyclotomic coset $S_q(a)$ is said to be of

type I if $S_q(a) = S_q(-a)$ or **type I'** if $S_q(a) \neq S_q(-a)$.

In addition, if $q = p^3$ is a prime-cubed number, a q -cyclotomic coset $S_q(a)$ is said to be of

type II if $S_q(a) = S_q(-pa)$ or **type II'** if $S_q(a) \neq S_q(-pa)$ and

type III if $S_q(a) = S_q(-p^2a)$ or **type III'** if $S_q(a) \neq S_q(-p^2a)$.

Definition 3 For $a \in \mathbb{Z}_m$, denote by $\text{ord}(a)$ the **additive order** of a in $\mathbb{Z}_m$ and when $\gcd(a, m) = 1$, denote by $\text{ord}_m(a)$ the **multiplicative order** of a .

Definition 4 The **Euler's totient function** ϕ is a function defined by

$$\phi(n) := |\{k \mid 1 \leq k \leq n \text{ and } \gcd(n, k) = 1\}|$$

for all positive integers n .

RESULTS

Relations between q -cyclotomic cosets of type II and III are proven.

Theorem 1 Let q be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. For each $a \in \mathbb{Z}_m$, $S_q(a)$ is of type II if and only if $S_q(a)$ is of type III.

Some relations between q -cyclotomic cosets of types I, II and III are shown.

Theorem 2 Let q be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. For each $a \in \mathbb{Z}_m$. If $S_q(a)$ is of type II or III, then it is of type I.

Characterizations of q -cyclotomic cosets of type II and also type III are shown.

Theorem 3 Let $q = p^3$ be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. For each $a \in \mathbb{Z}_m$, the q -cyclotomic coset $S_q(a)$ is of type II if and only if $\text{ord}(a) \mid p^{3s+1} + 1$ for some integer $s \geq 0$.

The enumerataion q -cyclotomic cosets of type I is given in [1]. Enumerations of q -cyclotomic cosets of type II and type III are shown in the following theorem.

Theorem 4 Let q be a prime-cubed integer and let m be a positive integer such that $\gcd(m, q) = 1$. Then the number of q -cyclotomic cosets of type II and type III is

$$\sum_{d \mid m} \lambda_p(d) \frac{\phi(d)}{\text{ord}_d(q)},$$

where $\lambda_p : \mathbb{N} \rightarrow \{0, 1\}$ is a function defined by

$$\lambda_p(t) = \begin{cases} 1 & \text{if there exists an integer } s \geq 0 \text{ such that } t \mid p^{3s+1} + 1, \\ 0 & \text{otherwise.} \end{cases}$$

The results for type I', II', and III' can be obtained as the complement of type I, II, and III.

Lastly, one of the number theoretical results are shown.

Corollary 5 Let a be a positive integer and p be a prime number. The following statements are equivalent:

- (1) $a \mid p^{3s-2} + 1$ for some integer $s \geq 0$.
- (2) $a \mid p^{3s-1} + 1$ for some integer $s \geq 0$.
- (3) $a \mid p^{3s+1} + 1$ for some integer $s \geq 0$.
- (4) $a \mid p^{3s+2} + 1$ for some integer $s \geq 0$.

In general, the study of different types of q -cyclotomic cosets in $\mathbb{Z}_m$ where $q = p^k$, p is a prime and k is a positive integer is interesting.

EXAMPLE

A 2^3 -cyclotomic coset of $\mathbb{Z}_{45}$ containing a is the set $S_8(a) = \{8^i a \mid i = 0, 1, \dots\}$. All cosets are $S_8(0) = \{0\}$,

$$S_8(1) = S_8(8) = S_8(17) = S_8(19) = \{1, 8, 17, 19\},$$

$$S_8(2) = S_8(16) = S_8(34) = S_8(38) = \{2, 16, 34, 38\},$$

$$S_8(3) = S_8(6) = S_8(12) = S_8(24) = \{3, 6, 12, 24\},$$

$$S_8(4) = S_8(23) = S_8(31) = S_8(32) = \{4, 23, 31, 32\},$$

$$S_8(5) = S_8(40) = \{5, 40\},$$

$$S_8(7) = S_8(11) = S_8(29) = S_8(43) = \{7, 11, 29, 43\},$$

$$S_8(9) = S_8(18) = S_8(27) = S_8(36) = \{9, 18, 27, 36\},$$

$$S_8(10) = S_8(35) = \{10, 35\},$$

$$S_8(13) = S_8(14) = S_8(22) = S_8(41) = \{13, 14, 22, 41\},$$

$$S_8(15) = S_8(30) = \{15, 30\},$$

$$S_8(20) = S_8(25) = \{20, 25\},$$

$$S_8(21) = S_8(33) = S_8(39) = S_8(42) = \{21, 33, 39, 42\},$$

$$S_8(26) = S_8(28) = S_8(37) = S_8(44) = \{26, 28, 37, 44\}.$$

The above cyclotomic cosets have different types, for example,

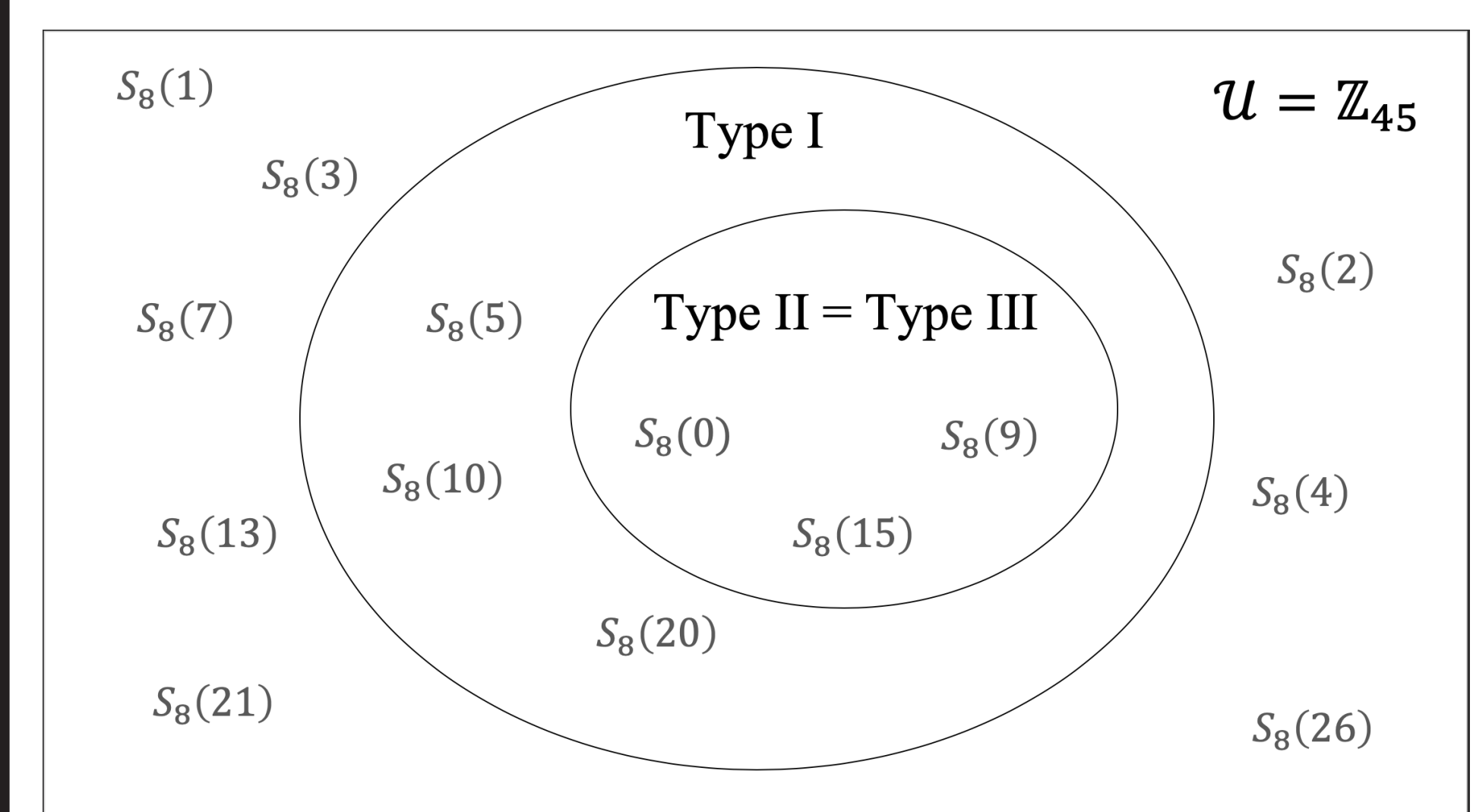
$S_8(9)$ is of type I, II and III

$S_8(5)$ is of type I, II' and III'

$S_8(1)$ is of type I', II' and III'.

EXAMPLE

From Theorem 1, Theorem 2, and the above example. We have the following Venn diagram.



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